ResearchAgent



Research with a wallet.

ResearchAgent answers from free sources first, then pays writers for the articles it is still missing, **over x402 on the XRP Ledger**, inside a budget you set. The writer is paid only by publishing the article's key on the ledger.

Try it live	https://ra-front-651013803640.asia-southeast1.run.app · Shared XRPL Testnet sandbox · no real value · open 13–19 Nov
Watch a run	collaboration95.github.io/theFastandtheFungible/demo/ · a recorded Testnet run, any time

```
$ npx -y https://github.com/Collaboration95/theFastandtheFungible/releases/download/v1.1.1/researchagent-mcp-1.1.1.tgz try --testnet
```

- Testnet · test funds · fictional demo writers, plus our own labelled team-authored notes · TEST-SGD stands in for RLUSD

From a hackathon prototype to a product anyone can check on the ledger

At SingHacks a language model proposed each purchase and a person approved every quote before payment; payment was a fixture by default, with optional Testnet settlement.

	At SingHacks	Now
Who decides to buy	A language model proposed; a person approved every quote	Policy code buys within your budget after one plan confirmation; a decision model (fixed rules in the sandbox) only scores
Fair exchange	Pay first, then read	Key-reveal escrow: the writer is paid only by publishing the article's key
Refunds	None	A broken promise is refunded on the ledger
Proof	Citations checked inside the app	An offline verifier anyone can run, in TypeScript or Python
Ways in	One web page	A live sandbox, an MCP server for Claude Code and Codex, SDKs and a Mac app

270
pull requests merged since 21 Sep

15
releases, v0.1.0 to v1.1.1

61
Testnet transactions you can re-check live on our ledger page

224
receipt test vectors; two verifiers, written separately, agree on all

Counted at bd52beff on 11 Nov 2026 (git and gh queries in site/progress/data.json) · "At SingHacks" is the 19–20 Sep event build (e16435a3)

One line runs the whole money path on real Testnet

```
$ npx -y https://github.com/Collaboration95/theFastandtheFungible/releases/download/v1.1.1/researchagent-mcp-1.1.1.tgz try --testnet
```

```
Testnet · no real value · Fixture models · fictional and team-authored writers
1. Create: 4AF8D55C…452022B6
2. Finish · sha256(K) = Condition ✓: 47FB0F5C…E9E83FA6
3. Refund · writer-signed · fixture writer cooperates (a real writer could refuse): 6A449846…702CE755
4. Re-buy create: DF565032…4E4EF373
5. Re-buy finish · sha256(K) = Condition ✓: F1AA7E20…5DB3A95F
6. Anchor: A722EB98…6FACB8AF
VERIFY 7cb2a7a4-2f7b-40d0-a84d-f5dbbf65fb98
PASS CONTRIBUTIONS 1/1
PASS MANIFEST   7/7
PASS PASSAGES   7/7
PASS CITATION   20/20
PASS GRANT      1/1
PASS RECEIPT    3/3
PASS REFUND     2/2
PASS AUDIT CHAIN 1/1
PASS testnet 12/12 (read-only, s.altnet)
Skipped writer credentials: the writer's identity is self-declared; checking it needs an online lookup
Skipped historical origin: not needed for this receipt format (one origin per article)
PASS audit anchor: Validated XRPL Testnet ledger 22238845
Skipped is not failed. PASS does not prove who a writer is.
Finished in 59.2 s · budget S$2.00 · net S$0.40 · gross S$0.70
RESULT PASS
```

84.2 s

from an empty npm cache to six Testnet links and RESULT PASS: buy, refund a broken promise, re-buy, anchor, verify. One cold run on one Mac.

2 of 4

cold runs overran the 90 s target. We keep the misses.

If the faucet fails, it prints a labelled SIMULATED run and the reason. The output here is a committed fresh-state run; its own clock excludes the npm download. Links are shortened to their hashes.

— What we built

One question, one budget, and only code can spend it

1
Ask
A question and a budget you set.

2
Confirm the plan
A short sheet confirms the plan, not a purchase.

3
Search writers
Writers return abstracts and signed promises, never premium text.

4
Decide
A decision model (fixed rules in the live sandbox) scores what is worth buying, weighted by trust.

5
Pay
Policy code locks payment in a key-reveal escrow on XRPL.

6
Verify and answer
Decrypt, check every promise, refund a broken one, cite, anchor.

Five hard gates, on every run

1. No premium bytes before a grant.
2. The budget is the only spending authority; only policy code can buy, never the language model.
3. One charge per intent, across retries and restarts.
4. Every citation points at a delivered passage.
5. Anything simulated or substituted is labelled.

0 gate violations

across the v5 held-out and off-topic question banks on Testnet (11 Nov).

Spending caps for MCP clients default to zero, and the calling model cannot raise them.

The writer is paid only by publishing the key: delivery and payment are one ledger event

Writer

Encrypt and promise

A fresh key K per buyer. Signs the quote: `sha256(K)`, ciphertext hash, passage manifest, price.

Buyer policy

EscrowCreate

Native XRPL Escrow. Condition = `sha256(K)`, with a CancelAfter deadline.

Writer

EscrowFinish

To be paid, the writer must reveal K as the fulfilment. K is now public on the ledger.

Buyer policy

Decrypt and check

Decrypt, check every promised passage, then grant and cite.

Promise kept	Create →	Finish, K revealed →	decrypt ✓ passages ✓ →	cited	`2F5C3D75…` → `528DFA6F…`
Promise broken	Create →	Finish, K revealed →	claim missing →	writer-signed refund + fraud proof	`E95BED8E…` → `B248161A…`
Writer no-show	Create →	no key by CancelAfter →	EscrowCancel: money back, nothing granted		`8DA4ECE7…` → `F89DD9B1…`

> **What the ledger does and doesn't enforce.** Key reveal makes **delivery** atomic: no key, no payment. **Quality** rests on the signed promises, a fraud proof anyone can check, and reputation. After a finish, the refund is writer-signed: our fixture writer cooperates, and a real writer could refuse. No smart contract: Escrow with a PREIMAGE-SHA-256 condition is native XRPL.

The same key-reveal escrow, settled in a stablecoin-style token

TEST-SGD through XLS-85 token escrow

CLI and evidence: the live sandbox pays in Testnet XRP.

- **Opt in** with `--iou` on the MCP server or the web app.
- **Preflight refuses** a missing issuer locking permission, frozen or unauthorized trust lines, short balances and transfer fees.
- **Both verifiers** check currency, issuer and the writer's trust-line balance change, because EscrowFinish has no delivered_amount.

`create E36AF9F1…18878C5C · finish 384CA34F…181C02F5 · anchor 8174E3BE…8ADC35A3`

Official RLUSD: one issuer flag away

- **Payments work.** Official Testnet RLUSD paid a writer on 3 Nov (`1F2449A4…`), and a broken promise was refunded.
- **Escrow is blocked by the issuer, not the network.** Token escrow is enabled on Testnet and Mainnet. Only the RLUSD issuer's flags block it, on both: AllowTrustLineLocking is off. One AccountSet by the issuer would allow it.
- **So we escrow our own token** and say so on every amount, receipt and ledger row. It is our top builder-feedback finding.

● TEST-SGD · an IOU from our Testnet issuer · no real value · stand-in for RLUSD · no peg, backing or mainnet claim

— Technical depth

Don’t trust our agent. Verify it, against the ledger.

```
# Python, no Node: a receipt the live sandbox exported
$ pip install …/v1.1.1/researchagent-1.1.1-py3-none-any.whl
$ researchagent verify receipts.json --ledger
PASS checks: 46 passed, 0 failed, 2 skipped
PASS audit anchor: Validated XRPL Testnet ledger 22228575
PASS testnet 11/11 (read-only, s.altnet)
Skipped is not failed. PASS does not prove who a writer is.
RESULT PASS
```

Any receipt the app or the live sandbox exports can be checked the same way, by anyone, offline or against the ledger, with release v1.1.1.

- **Two verifiers, written separately**: TypeScript, and Python’s standard library. They agree on all 224 shared vectors: 19 must pass and 205 tampered ones must fail.
- **61 Testnet transactions** are re-checked live, read-only, every time our public ledger page loads.
- **Every finished answer is anchored:** an AccountSet whose memo is the audit-chain head, even when nothing was bought.
- **Persist before submit.** A restart reconciles the same signed transaction and never signs twice; a missing one is *pending*, not failed.
- **Exact delivery.** Partial payments are refused before submit; amount, currency and issuer are checked after.
- **Prompt injection can’t spend:** model inputs are untrusted, and only policy code can buy.

Output: docs/submission/evidence/v1.1.1/ · v1.1.1 wheel, committed hosted export, 11 Nov · tests/fixtures/receipt-conformance/ · the /ledger/ page

— Reachability

Every way in runs from one package, for people and for other agents

In the browser

The live sandbox: ask your own question, nothing to install.

Shared XRPL Testnet sandbox · no real value · open 13–19 Nov

Claude Code, any MCP client

`claude mcp add … --testnet`. Spending caps default to zero; the calling model can't raise them.

Cold start 6.3 s · 3.4 MB package

Local app, no clone

`app --testnet` serves the real app on 127.0.0.1 with faucet wallets.

Ready in 27.0 s from an empty cache

TypeScript and Python SDKs

Ask, stream, verify. No purchase method: only backend policy can buy.

The Python verifier needs no Node

Writers in one command

`publish ./notes --testnet`: faucet wallet, XRPL Credential, key-reveal escrow.

33.9 s from `init` to a validated Credential

Hire the agent over x402

Another agent pays ResearchAgent; it buys, pays every cited writer a royalty and returns the change. Run on Testnet.

One line: `try --testnet --hire`

Does it buy the right thing? v5 blind held-out: 16/20 bought and cited, 0/10 off-topic purchases (run once, 11 Nov, on XRPL Testnet: fixture research and decision models with answers written offline, team-authored writers). The key fact leads 7/20 answers. Packages: GitHub Releases; not yet on npm, PyPI or the MCP Registry.

— Feasibility

Fractions of a cent per purchase, and it recovers from crashes

435 drops

network fees per key-reveal purchase

12 paid by the buyer to lock the escrow and 423 by the writer to reveal the key, because the finish carries it: 0.000435 XRP as we sign on Testnet (rippled's own minimum is lower).

0 double charges

under chaos, across retries and restarts

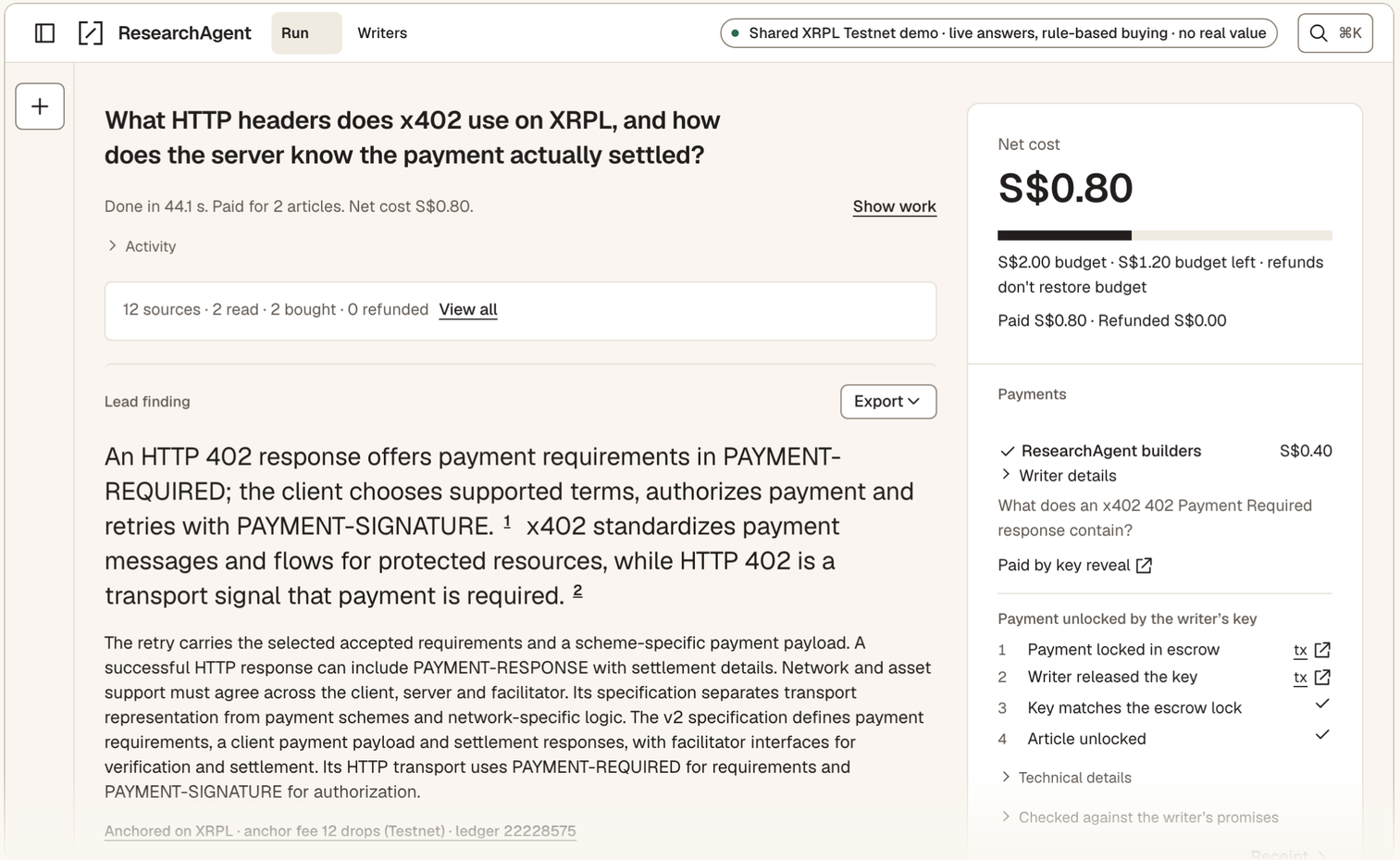
500 seeded runs with 130 real process kills; the reconciler recovers in-flight payments. Synthetic ledger, 3 Nov.

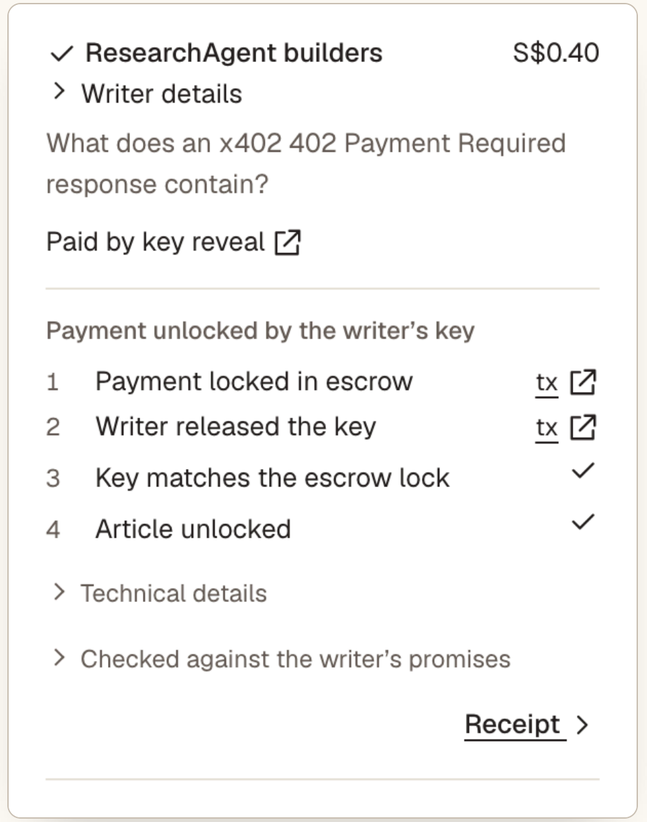
How it runs

- **Hosted for judges on Cloud Run,** 13–19 Nov: one shared Testnet payer, caps per visitor, owner controls off.
- **Persist before submit:** a restart never signs twice; a missing transaction is pending, never failed.
- **Each answer names its models:** live, or a labelled fixture fallback.
- **Mainnet is disabled in code.** What it needs is on the compliance slide and in our builder feedback.

docs/evidence/round3-a/probe.json · docs/evidence/chaos-279-500.json · README · Status and limits

The money is the hero, and every label tells the truth





One bought article: paid by key reveal.

One environment indicator in the header · **the answer leads** · **each payment step** links to its transaction

One live run on the hosted demo, 11 Nov (docs/evidence/hosted-export-nov11/) · XRPL Testnet · live answers, rule-based buying, team-authored writers · captured by hosted-shots.mjs, unedited

— Reachability · Feasibility

Agents need facts behind paywalls, and writers need to be paid per read

Who pays

- **The asker,** a person or their agent, pays per article, inside a budget they set per question.
- **The writer** sets a price per article and is paid on the ledger, with an XRPL address and no card processor.
- **We earn a service fee** when another agent hires ResearchAgent over x402; part of it goes to every cited writer.

Market

- **Writers are being left out.** Small publishers have lost search referrals, and AI licensing deals mostly go to major brands.
- **Today's tools price a whole site** per crawl or by revenue share.
- **None we reviewed** judges value before buying, checks delivery against a promise, or refunds.

Why now

- **Agents act through tools:** Claude Code, Codex and other MCP clients call them on their own.
- **x402 gives HTTP a payment step** a machine can complete.
- **XRPL token escrow** is enabled on Testnet and Mainnet, so the same escrow can settle in a stablecoin once issuers allow it.

> **Where we are:** no customers, pilots or revenue yet, and we have not sized the market. Writer number one is us, labelled; the next are people who consent. The service fee and royalty split ran once on Testnet.

Open Markets / CJL, "Same Gatekeepers, New Tollbooths" (Apr 2026) · Cloudflare Pay Per Crawl and x402 posts · docs/FINAL-PUSH.md §14 · docs/evidence/hire-274.md · qa-prep Q4

— Reachability · compliance readiness

Built for audit. Honest about what a regulated launch still needs.

Every spend is authorised by deterministic policy code, capped, and written to a hash-chained audit log whose head is anchored on the XRP Ledger, even when the agent refuses to buy. It is Testnet only, and each gap has a named next step.

	Built (Testnet)	Not done
Custody	Keys stay server-side, out of logs, audit rows and the browser	KMS or HSM and rotation; the hire flow holds funds in transit
Spend limits	A budget per question and a daily cap; only policy code buys	Maker-checker, velocity limits
Audit	Hash-chained log, anchored heads, offline verification	Signed, durable retention; retention periods undecided
Screening, KYB	A payee denylist hook; XRPL Credentials badge for writers (publish CLI; none in the live sandbox)	No sanctions provider, no KYB, no Permissioned Domains
Regulation	Questions mapped for MAS (PSA, DPT services), GENIUS Act, Travel Rule	No licence, no legal opinion; mainnet blocked in code

Ranked builder feedback for Ripple, each with a runnable repro

1 **Issuer permission blocks RLUSD escrow**
Token escrow is enabled; the issuer lacks AllowTrustLineLocking. Fix: a network and issuer preflight.

2 **An issued amount must mean the delivered amount**
Only validated delivered_amount proves what arrived. Fix: negative issued-asset test vectors.

3 **A signed IOU quote overflowed the 1 KiB Memo**
Observed on Testnet, 7 Nov. Fix: check the serialized Memo size before signing.

4 **Escrow expiry doesn’t return funds by itself**
Funds stayed locked until an explicit EscrowCancel. Fix: document expiry versus cancellation.

5 **Endpoints disagree on retained history**
One answered “not found” for a validated hash; HTTP lacks browser CORS. Fix: an endpoint matrix.

6 **Settled invoice consumption blocks recovery**
A same-payment retry is refused while delivery still needs recovering. Fix: idempotent replay.

7 **A header-only x402 v2 challenge is not retried**
The client returns the unpaid 402. Fix: read the v2 header first.

\+ **Mainnet readiness: a checklist, not a claim**
Plus what worked well: escrow, delivered_amount, Credentials, fast finality.

> 7 ranked findings with runnable repros · 35 numbered findings (B01–B35) · 15 submissions on 4–5 Sep · 8 upstream drafts, 0 filed.

— Team · check every claim

A small team, a fleet of coding agents, and a receipt for every payment

Guruprasath

Product and engineering since SingHacks: plans the work, directs the coding agents and owns the merge gate.

Miul

SingHacks build: the research-plan contracts, quote binding and the first test harness.

Andrew Nguyen

Before SingHacks: the idea briefs and data sources.

How we build

Coding agents work in parallel lanes; every merge passes one local gate of tests and the five hard gates. 270 pull requests since 21 Sep.

Check it yourself

Live sandbox, 13–19 Nov	`ra-front-651013803640.asia-southeast1.run.app`
Recorded Testnet run	`collaboration95.github.io/…/demo/`
Ledger page, live checks	`collaboration95.github.io/…/ledger/`
Progress since SingHacks	`collaboration95.github.io/…/progress/`
Release, every artifact	`v1.1.1 on GitHub Releases`
Code and evidence	`github.com/Collaboration95/theFastandtheFungible`
Demo video	linked in the submission form

People and roles from the git history (git shortlog) · AGENTS.md (how work is run) · Testnet XRP and TEST-SGD have no value; receipts keep ledger indexes if Testnet resets